COMPETENCE BASED CHEM QUESTIONS

CBCQ



QUESTIONS 2025

Specific Competence Assessment Tools

FORM ONE 2025 SCAT- 1

F1 -MIDTERM-1 EXAMS QUESTIONS.

The basic concept of Chemistry

1. Use the knowledge of application and branches of chemistry to answer the following questions.
    a. Complete the following table

EXPLANATION	APPLICATION	EXAMPLE
	Baking	
It involves fractional distilled of crude oil into useful portions		
	Medicine	Paracetamol Aspirin
Is a chemical process that involves interaction of water, soap and dirt		
	Biochemistry	Carbohydrates, proteins and vitamins.
If the body is not functioning normally, it is said to be sick/ill. Laboratory test are done to diagnose the illness/sickness.		Malaria parasites HIV test
	Argo-chemistry	Plastics, glass, fertilizers, insecticides, soaps, cements

    b. List four branches of chemistry
    c. Name three main branches of science
    d. Name three career when study chemistry

2. Use your comprehension knowledge of disciplines related to chemistry to answer the following
    a. Complete the following table

EXPLANATION	**SCIENTIST**	**FIELD**	**CHEMISTRY DISCIPLINES**	**EXAMPLE**
The field that helps to enhance the agricultural products			Agro-chemistry	Fertilizers Pesticides
	Geologist	Geology	Geochemistry	
	Biochemist	Biology	Biochemistry	

    b. Define the term drug.
    c. A patient was given tablets with prescription 2 x 3 on the envelope. Clearly outline how the patient should take the tablets.
    d. Identify the science field (a) above.

3. Use the knowledge of application of chemistry in our societies to attempt the following questions
    a. Complete the following table

FIELD	DIAGRAMS	EXPLANATION
		
		
		
		

    b. Define the term prescription
    c. A form one student went to the school clinic and was prescribed malarial drug to take 2 x 3. Explain how the student was supposed to take the drugs
    d. Supposing the student took the drugs at 7:00am in the morning.
        i. Calculate the other hours of the day when he is expected to take other drugs. Assuming that to maintain consistent therapeutic level need 8-hour intervals.
        ii. Identify this science fields.

Laboratory technique and safety

4. Use your comprehension knowledge in laboratory safety to
    a. State the laboratory rules that should be applied to prevent the following accidents
        i. Mistaking hydrochloric acid to be distilled water
        ii. A student got burnt after secretly lighting up a magnesium ribbon
        iii. A student got severe stomach ache after eating some bread during chemistry laboratory session
    b. State two laboratory rules to observe when preparing a poisonous gas
    c. State what students should do in case of a major accident such as fire outbreak in the chemistry laboratory
    d. State the two causes of accidents in a chemistry laboratory
    e. Putting off flames after use is one of the safety rules in the laboratory, state other three safety rules.

5. Use your knowledge on first aid to briefly explain the following facts
    a. Mugure a form one student was stung by a wasp in the laboratory, the technician applied sodium hydroxide on her. Explain why this was done.
    b. It is not allowed for a snake bite person to take alcohol. Why?
    c. For insect bite person, first aider should not apply compress finger to sting. Why?
    d. Corrosive poisoning person should not induce vomiting. Why?
    e. It is allowed for corrosive poisoning victim to drink milk or water. Explain significant role why this was applied?

6. Briefly attempt the following from the concept of first aid
    a. What is first aid kit
    b. Complete the following table appropriate

NAME	DIAGRAM	ITERM USES
		
		
		
		

    c. Identify the first aid chemical items used when victim have the following condition
        i. Dehydration due to vomiting
        ii. Excessive bleeding of small cut
        iii. Bite from insect or snake
        iv. Stiffness of the muscle
    d. A form one student was bite with snake during laboratory session, the technician then applies an antibiotic ointment on wound. Explain why this item was used?

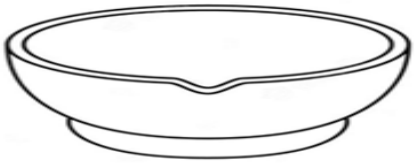
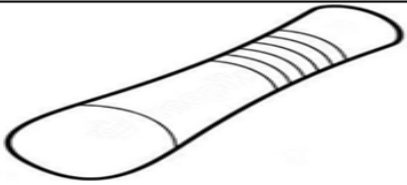
Basic chemistry laboratory apparatus

7. Use your comprehension knowledge of laboratory apparatus to attempt the following
    a. Complete the following table

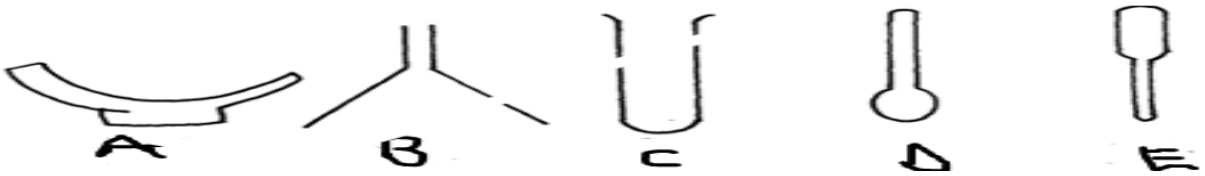
Apparatus	Use
	1mk
	1mk

    b. State two apparatus used as source of heat in a laboratory
    c. State the function of the following apparatus in the laboratory
        i. Round bottomed flask
        ii. Evaporating dish
    d. Name the apparatus used to measure the following
        i. Accurate volume of liquids (any three)
        ii. Amount of solid
        iii. Temperature of boiling water
    e. Draw two separate diagrams to differentiate a flat-bottomed flask of 250ml and volumetric flask of 250ml

8. Apply the knowledge of laboratory apparatus to the following questions
    a. Complete the following table appropriate

Diagram	Name	Use
		
		

b. Volumetric flask and flat-bottomed flask are made of glass. Explain why? (Any 4)
c. List three scientific skills can be acquired when performing chemistry experiments.
d. Use the diagram below to attempt the following questions



i. Give the correct order of the apparatus, using the letters only, to show the correct arrangement that can be used to prepare sample and investigate the nature of PH of a sample of onion solution.
ii. Name the apparatus above and chemical substance added in this experiment.

e. Complete the following table

APPARATUS	NAME
	
	

Chemical safety symbols

9. The following symbols had been put on containers with certain chemicals, identify the symbol and explain there meaning
    a. Complete the following table appropriate

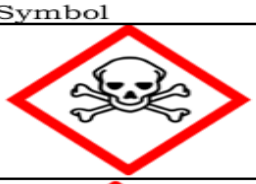
NAME	SYMBOLS	EXPLANATION
		
		
		

    b. Identify the hazard symbols from the following explanation
        i. Substance that increases the rate of combustion
        ii. Substance that emits harmful radiation such as alpha, beta, gamma or neutrons
        iii. Substance that causes body wounds or burn various thing such as clothes
        iv. Substance which should never be brought near open flame since they can catch fire easily
    c. Explain and draw the following warning sing and provide significant example each
        i. Radioactive
        ii. Electrical hazard
    d. Complete the following table appropriate

WARNING SING	EXAMPLE	EXPLANATION
		
		

10. The following are hazard symbols were observed by a student in one stock bottle in chemistry laboratory. Identify the symbols

a. Complete the following table

	Symbol	
A		1mk
B		1mk

b. Identify the warning sing from the following chemical substance
   i. Potassium permanganate
   ii. Uranium
   iii. Sulphuric acid
   iv. Petrol

c. Give the example of the following warning sing
   i. Oxidant
   ii. Toxic

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Prefer Calling Group of Examiners @ 0738380920 for Marking Schemes and the rest of semester question 2025

